Neutrino Mass Hierarchy from the $GF(27)^*$ Frobenius Structure

$$\Delta m^2_{\mathrm{atm}} = 120\, m_\nu^2, \quad \Delta m^2_{\mathrm{sol}} = \frac{11}{3}\, m_\nu^2$$

Johann Pascher
ORCID 0009-0000-6518-4064

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Abstract

Starting from the Frobenius decomposition $GF(27)^* \cong \mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$ (Doc. 339), the measured neutrino mass differences Δm^2_{atm} and Δm^2_{sol} are derived from algebraic properties of the four Frobenius orbits in $\mathbb{Z}_{13}$. The atmospheric mass difference $\Delta m^2_{\text{atm}} = (11^2 - 1)m_\nu^2 = 120\, m_\nu^2$ (1.0 % deviation) and the solar $\Delta m^2_{\text{sol}} = (|\mathbb{Z}_{13}| - 2)/3 \cdot m_\nu^2 = (11/3)m_\nu^2$ (0.46 % deviation) follow from the number 11, the maximum element of the fourth Frobenius orbit $\{7, 8, 11\}$ in $\mathbb{Z}_{13}$. Orbit 3 $\{4, 10, 12\}$ is algebraically self-inverse and carries Majorana character. The coherent venting amplitude $p_{\text{vent}} = 5/24$ is proved algebraically from the branching rule $\rho_4 \downarrow_{A_4} = \rho_3^{A_4} \oplus \rho_1^{A_4}$ and the index $|A_5 : A_4| = 5$. The SICC venting formula is equivalent to the standard oscillation formula. Status: **[B]** for the group theory; **[K]** for the mass differences.

.1 Starting Point

In Doc. 339 the Frobenius decomposition $GF(27)^* \cong \mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$ is established **[B]**. The four three-element orbits of $\mathbb{Z}_{13}$ under the Frobenius $\phi : x \mapsto 3x$ are:

Orbit	Elements	Sum	Product mod 13
1	$\{1, 3, 9\}$	13	1
2	$\{2, 5, 6\}$	13	8
3	$\{4, 10, 12\}$	26	12
4	$\{7, 8, 11\}$	26	5

The FFGFT neutrino base mass is speculatively set as $m_\nu = \xi^2/2 \cdot m_e \approx 4.54$ meV **[S]** (Doc. 047). The present calculations test whether the measured mass differences follow from the Galois structure.

.2 Algebraic Properties of the Orbits [B]

Orbit 4 as a multiple of Orbit 1

$$\{7, 8, 11\} = 7 \cdot \{1, 3, 9\} \pmod{13},$$

since $7 \cdot 1 = 7$, $7 \cdot 3 = 21 \equiv 8$, $7 \cdot 9 = 63 \equiv 11$ (mod 13). Since $2^{-1} \equiv 7 \pmod{13}$:

$$\boxed{\text{Orbit}_4 = 2^{-1} \cdot \text{Orbit}_1 \pmod{13}.}$$

Duality Orbit 2 ↔ Orbit 4

The minimal elements and their multiplicative inverses in $\mathbb{Z}_{13}$:

Orbit	min	$\min^{-1} \pmod{13}$	lies in
1	1	1	Orbit 1
2	2	7	Orbit 4
3	4	10	Orbit 3
4	7	2	Orbit 2

Orbits 2 and 4 are algebraically inverse: $2^{-1} = 7$, $7^{-1} = 2$. Orbit 3 is self-inverse: $4^{-1} = 10$, $10^{-1} = 4$, $12^{-1} = 12$ (mod 13). The self-inverse nature of Orbit 3 corresponds algebraically to Majorana character. The sum: $4 + 10 + 12 = 26 = |GF(27)^*|$.

.3 Neutrino Masses from the Orbit Structure [K]

The factor 11

The maximum element of the fourth orbit, $\max(\{7, 8, 11\}) = 11$, follows from:

$$11 = |\mathbb{Z}_{13}| - |\text{Frobenius fixed points in } \mathbb{Z}_{13}| - 1 = 13 - 1 - 1 = 11.$$

This factor determines the mass hierarchy (normal ordering):

State	Galois origin	Mass	Value
ν_1	Orbit 1 (D4, 4D)	m_ν	4.54 meV
ν_2	Orbit 3 (self-inverse, E6, solar)	$\sqrt{14/3}\,m_\nu$	9.81 meV
ν_3	Orbit 4 (E8, atm., heaviest)	$11\,m_\nu$	49.96 meV

The sum $\sum m_i = (1+\sqrt{14/3}+11)\,m_\nu \approx 64.3$ meV is consistent with the Planck-2018 bound $\sum m_i < 120$ meV.

Atmospheric mass difference

$$\Delta m^2_{\text{atm}} = m^2_{\nu_3} - m^2_{\nu_1} = (11^2-1)\,m_\nu^2 = 120\,m_\nu^2 = 2.476\times10^{-3}\ \text{eV}^2.$$

Measured: 2.500×10^{-3} eV^2. **Deviation: 1.0 %.**

The factor $120 = |A_5\times\mathbb{Z}_2| = 11^2-1$ is the order of the full icosahedral symmetry group – consistent with the A_5 symmetry of the HLV sector (Doc. 285).

Solar mass difference

$$\Delta m^2_{\text{sol}} = m^2_{\nu_2} - m^2_{\nu_1} = \left(\frac{14}{3}-1\right)m_\nu^2 = \frac{11}{3}\,m_\nu^2 = 7.565\times10^{-5}\ \text{eV}^2.$$

Measured: 7.530×10^{-5} eV^2. **Deviation: 0.46 %.**

The algebraic derivation of $11/3$ **[B]**:

$$\frac{11}{3} = \frac{|\mathbb{Z}_{13}| - |\text{fixed points of } GF(27)^*|}{|\text{Frobenius order}|} = \frac{13-2}{3}.$$

Numerator: $13-2=11$ (the two fixed points $\{+1,-1\}$ of $GF(27)^*$ subtracted from $|\mathbb{Z}_{13}|$). Denominator: 3 (the order of $\phi : x\mapsto x^3$).

.4 Kappa and the SICC Connection [K]

Relation κ – Δm^2_{atm}

From Doc. 339 (and Paul Masic's SICC derivation):

$$\kappa = \frac{8}{r_e}\cdot m_e\,\xi^2 = 6\,\xi^2\,m_e = 12\,m_\nu.$$

It then follows immediately:

$$\Delta m^2_{\text{atm}} = (11^2-1)\,m_\nu^2 = 120\,m_\nu^2 = \frac{5}{6}\,\kappa^2.$$

The ratio:

$$\frac{\kappa^2}{\Delta m^2_{\text{atm}}} = \frac{144}{120} = \frac{6}{5}.$$

The coherent venting amplitude $p_{\text{vent}} = 5/24$ [B]

If W-venting events accumulate as coherent quantum phases rather than energy loss, the oscillation phase reads:

$$\phi_{\text{venting}} = p_{\text{vent}} \cdot \frac{\kappa^2 L}{\hbar c E_\nu}.$$

For $\phi_{\text{venting}} = \Delta m^2_{\text{atm}} \cdot L/(\hbar c E_\nu)$, one requires:

$$p_{\text{vent}} = \frac{\Delta m^2_{\text{atm}}}{4\kappa^2} = \frac{120\, m_\nu^2}{4 \cdot 144\, m_\nu^2} = \frac{120}{576} = \frac{5}{24}.$$

Algebraic derivation of $p = 5/24$ from the group structure [B]

The branching rule for restricting the 4D representation ρ_4 of A_5 to the subgroup A_4 (tetrahedral group, order 12) reads:

$$\boxed{\rho_4 \downarrow_{A_4} = \rho_3^{A_4} \oplus \rho_1^{A_4}.}$$

This follows from the A_4 character table: the characters of ρ_4 on the A_4 classes $(e, (123), (132), (12)(34))$ are $(4, 1, 1, 0)$, and the scalar products with A_4 characters yield multiplicity 1 for both $\rho_1^{A_4}$ and $\rho_3^{A_4}$.

Consequence: A_4 leaves a one-dimensional sub-representation (the W-direction) fixed in ρ_4. That is, A_4 is the *stabiliser* of the W-direction in A_5.

In A_5 there are exactly

$$|A_5 : A_4| = \frac{60}{12} = 5$$

conjugates of A_4 – corresponding to the 5 tetrahedra embedded in the icosahedron. Each conjugate stabilises a different W-direction. The W-venting selects exactly one of these 5 directions (the physical 4th spatial direction), hence:

$$\boxed{p_{\text{vent}} = \frac{|A_5 : A_4|}{\text{Kissing}(D_4)} = \frac{5}{24}.}$$

Equivalence to the standard oscillation formula

The complete SICC venting formula reads:

$$\phi_{\text{venting}} = \frac{5}{24} \cdot \frac{\kappa^2 L}{\hbar c E_\nu} = \frac{5}{24} \cdot \frac{144\, m_\nu^2 L}{\hbar c E_\nu} = \frac{120\, m_\nu^2 \cdot L}{4\, \hbar c E_\nu} = \frac{\Delta m^2_{\text{atm}} \cdot L}{4\, \hbar c E_\nu}.$$

This is *identical* to the standard oscillation formula. All parameters – κ, p_{vent}, Δm^2_{atm} – derive from the Galois structure $GF(27)^*$ and the group structure $A_5 \supset A_4$. No free parameter, no DUNE calibration.

The complete derivation chain:

$$GF(27)^* \xrightarrow{[B]} m_\nu \xrightarrow{[K]} \kappa = 12\, m_\nu \xrightarrow{[K]} \Delta m^2_{\text{atm}} = 120\, m_\nu^2 \xrightarrow{[B]} p_{\text{vent}} = \frac{|A_5 : A_4|}{\text{Kissing}(D_4)} = \frac{5}{24}.$$

Non-closure of the angular structure: $\mathbb{Z}_{13} \times \mathbb{Z}_{24}$ [B]

The SICC simulation quantises directions onto the 24 kissing vectors of D_4; the Frobenius structure supplies orbit phases $2\pi k/13$. The two angular lattices are coprime:

$$\text{gcd}(13, 24) = 1, \qquad \text{lcm}(13, 24) = 312.$$

The combined angular structure therefore closes only after 312 steps, not 24 – a discrete spiral. This is the non-closure structure of Doc. 193 in angular form: there at the level of mass products $r_i \cdot r_j$, here at the level of $\mathbb{Z}_{13} \times \mathbb{Z}_{24}$.

Continuously, the spiral also appears across the beam spectrum: $\phi(E) \propto 1/E$ is not periodic in E. Over $[0.5,\ 5]$ GeV (DUNE) the phase winds through 1.155 turns with remainder 55.7° mod 2π – the curve does not close. This is invisible in a 1D phase histogram; in a polar plot (radius $=$ spectral weight, angle $= \phi$ mod 2π) the spiral becomes visible.

Fractal correction: not applicable to ratios

The fractal correction $K_{\text{frak}} = 1 - 100\,\xi$ (Doc. 011/133) applies to absolute masses, not to dimensionless ratios (Doc. 306). $\Delta m^2_{\text{atm}}/m^2_\nu = 120$ is a ratio; K_{frak} must not be applied here. As a check: applied, the deviation of Δm^2_{atm} would rise from $1.0\,\%$ to $2.3\,\%$; the effect on DUNE disappearance would be 0.2 percentage points – below detector resolution.

.5 Mixing Angles [K] / [S]

Two mixing angles can be read off from Orbit-2 elements with $< 6\,\%$ deviation **[K]**:

$$\sin^2\theta_{12} \approx \cos^2\!\left(\frac{2\pi\cdot 2}{13}\right) = 0.323 \quad \text{(measured: } 0.307,\ \text{dev. } 5.1\,\%)$$

$$\sin^2\theta_{23} \approx \cos^2\!\left(\frac{2\pi\cdot 5}{13}\right) = 0.560 \quad \text{(measured: } 0.545,\ \text{dev. } 2.8\,\%)$$

The reactor angle θ_{13} has two independent Galois derivations **[K]**:
(a) Galois formula from Doc. 328 (stronger):

$$\theta_{13} = \arcsin\!\big((25\,\xi)^{1/3}\big), \quad 25\,\xi = \frac{1}{300} = \frac{1}{5\cdot|A_5|},$$

$$\sin^2\theta_{13} = \left(\frac{1}{300}\right)^{2/3} \approx 0.02230 \quad \text{(measured: } 0.0220,\ \text{dev. } 1.4\,\%)$$

The factor $25 = 5^2$ comes from the Galois identity $43200 = |GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)| = 64 \cdot 25 \cdot 27$ (Doc. 338), where 5^2 is the square of the maximal A_5 representation dimension.
(b) Sum rule from R_ν (independent confirmation):

$$\sin^2\theta_{13} = \frac{2}{3}R_\nu = \frac{2}{3}\cdot\frac{11}{360} = \frac{11}{540} \approx 0.02037 \quad \text{(dev. } 7.4\,\%) \quad [1]$$

In Galois language: $2/3 = \min(\text{Orbit}_2)/\max(\text{Orbit}_1)$.

The same source independently gives:

$$\sin^2\theta_{12} = \frac{1}{3 - 2R_\nu} \approx 0.340 \quad \text{(dev. from Galois value } 0.323:\ 5.4\,\%)$$

The CP-violation phase $\delta_{CP} \approx -\pi/2$ (experimental hint) corresponds to Orbit-3 element $k = 10$: $\arg(e^{2\pi i\cdot 10/13}) = -83.1^\circ$, deviation 6.9° **[S]**.

Literature connection: T_{13} as a flavour symmetry group

The Frobenius group $T_{13} = \mathbb{Z}_{13} \rtimes \mathbb{Z}_3$ is known in the literature as a candidate for the flavour symmetry of neutrinos [2]. It is identical to the Frobenius structure derived here from $GF(27)^*$. The literature treated T_{13} as an abstract symmetry (without physical derivation); FFGFT provides the algebraic origin: $T_{13} = GF(27)^*$ under the Frobenius automorphism $\phi : x \mapsto x^3$.

.6 Orbit 3 as the Majorana Sector [B]

Orbit 3 $\{4, 10, 12\}$ is fully self-inverse in $\mathbb{Z}_{13}$:

$$4^{-1} \equiv 10, \quad 10^{-1} \equiv 4, \quad 12^{-1} \equiv 12 \pmod{13}.$$

The self-inverse nature means the associated particles are their own antiparticles – Majorana character. Additionally $4 + 10 + 12 = 26 = |GF(27)^*|$ **[B]**.

If sterile neutrinos exist, they lie algebraically in the Orbit-3 sector with minimal mass $m_{\text{sterile,min}} = 4\, m_\nu = 18.2\,\text{meV}$ **[S]**.

.7 Neutrinoless Double Beta Decay [K]

With measured PMNS mixing angles and Galois masses:

$$m_{ee} = \left|c_{12}^2 c_{13}^2\, m_1 + s_{12}^2 c_{13}^2\, m_2 + s_{13}^2\, m_3\right| \approx 7.1\,\text{meV} \quad \text{(Majorana phases zero)}.$$

The KamLAND-Zen bound $m_{ee} < 36\,\text{meV}$ is satisfied with large margin. The minimum (destructive interference): $m_{ee,\text{min}} \approx 1.2\,\text{meV}$.

.8 Conjugation Structure of the Orbits [B]

The four Frobenius orbits in $\mathbb{Z}_{13}$ form two conjugate $3/\bar{3}$ pairs under $x \mapsto -x \pmod{13}$, matching the conjugacy classes $C_{3_1}, C_{\bar{3}_1}, C_{3_2}, C_{\bar{3}_2}$ of T_{13} [2]:

$$-\text{Orbit}_1 = \text{Orbit}_3, \quad -\text{Orbit}_2 = \text{Orbit}_4.$$

Together with the inversion duality ($x \mapsto x^{-1}$, Doc. 339), the orbits carry a complete $\mathbb{Z}_2 \times \mathbb{Z}_3$ symmetry in $\mathbb{Z}_{13}$. Orbit 1 is also self-inverse under inversion ($3^{-1} = 9$, $9^{-1} = 3$, both in Orbit 1).

.9 Model Discrepancies for Context [S]

The A_4/TM_2 model [1] that provides the θ_{13} sum rule makes predictions that deviate from both the Galois picture and experiment:

- $\sin^2\theta_{23} = 0.433$–$0.441$ (lower octant) – measured: 0.545–0.560 (upper octant); Galois: 0.560 (upper octant).
- $\delta_{CP} \approx \pm 51^\circ$ – T2K hint and Galois ($k = 10$): $\approx -83^\circ$.
- $m_{ee} = 5.1$–$5.3\,\text{meV}$ (with Majorana phases) – Galois (zero phases): $7.1\,\text{meV}$.

For θ_{23} and δ_{CP}, the FFGFT Galois values agree better with experiment than the A_4 model. The θ_{13} sum rule is adopted; the remaining A_4 predictions are not.

.10 Falsifiable Predictions

1. **Neutrino masses (normal ordering):** $m_1 = 4.54$, $m_2 = 9.81$, $m_3 = 49.96$ meV, $\sum = 64.3$ meV. Testable: CMB-S4, Euclid (cosmology at 1 meV level). Independent convergence: Razzaghi et al. [1] extract from A_4 texture and experiment $m_2 = (9.21–9.92)$, $m_3 = (49.91–51.42)$, $\sum = (63.97–64.55)$ meV; our Galois values m_2, m_3 and $\sum$ lie within these ranges.
2. **Mass-difference ratio [K]**:

$$\frac{\Delta m^2_{\text{atm}}}{\Delta m^2_{\text{sol}}} = \frac{3(11^2-1)}{11} = 32.73 \quad \text{(measured: } 33.20).$$

   Once m_ν is measured to 1 meV precision, the ratio $\Delta m^2/m_\nu^2$ is directly testable.
3. **Orbit-3 sector (Majorana) [K]**: Sterile neutrinos (if they exist) are Majorana particles with $m_{\text{sterile,min}} = 4\,m_\nu = 18.2$ meV. Testable: IceCube, SBN (Fermilab), BEST experiment.
4. **Venting rate and DUNE observable [K]**: The coherent SICC venting amplitude $p_{\text{vent}} = 5/24 \approx 20.8\,\%$ per coherence step is a model-internal prediction. The DUNE observable depends on the beam spectrum: monochromatic 2.5 GeV gives $99.8\,\%$ ν_μ disappearance (first oscillation maximum, $\phi = 1.61$ rad); a Gaussian spectrum 2.5 ± 1.0 GeV clamped to $[0.5,\ 5]$ GeV gives an average of $76\,\%$ (SICC V3: $76.16\,\%$). For a direct prediction the real LBNF flux table must be used as weight.

.11 Open Points

1. The number $11 = |\mathbb{Z}_{13}| - 2$ governs both mass differences under the same algebraic principle **[K]**:
   - Solar derivation: $11/3 = (|\mathbb{Z}_{13}| - 2)/|\text{Frobenius order}| = (13-2)/3$
   - Atmospheric mass: $m_{\nu_3} = 11\,m_{\nu_1}$, since all 11 non-fixed elements of $\mathbb{Z}_{13}$ contribute to the dynamic mass oscillation (the 2 fixed points $\{+1, -1\}$ are invariant).

   Both derivations are unified under the same Galois principle.
2. Connection of the discrete orbit phases $2\pi k/13$ to the full PMNS U-matrix.

.12 Note on Authorship

This document was developed on 30 August 2026 in collaboration with the AI assistant Claude (Anthropic, Sonnet 4.6). The algebraic computations (Frobenius orbits in $\mathbb{Z}_{13}$, branching rule $\rho_4{\downarrow}_{A_4}$, character theory of A_5 and A_4, numerical verification of the mass differences) were carried out by the assistant and verified by the author. The physical interpretation, the connection to the SICC framework, and all status markers (**[B]**, **[K]**, **[S]**) are the author's responsibility.

.13 Notation

[ASSUMPTION]	Axiom or declared external input – not derived
[K]	Core – derived from ξ and the assumptions, numerically verified
[B]	Bridge – algebraically proved
[S]	Sketch – plausible, not yet fully worked out

.14 Verification Scripts

`pruef_340_neutrino_galois.py` (13/13 assertions passed);
`pruef_340b_spirale_breitband.py` (6/6 assertions passed: non-closure $\mathbb{Z}_{13} \times \mathbb{Z}_{24}$, spiral, broadband DUNE, K_{frak}).

Declared 30 August 2026, supplemented 31 August 2026 (θ_{13}, T_{13} literature, non-closure, broadband)

Bibliography

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[2] C. Hartmann, A. Zee, *Neutrino mixing and the Frobenius group T_{13}*, arXiv:1106.0333 [hep-ph] (2011).